

Final Electricity

Current $\rightarrow$ (+ve) to (-ve) flow
Charge $\rightarrow$ (-ve) to (+ve) flow

change, $q = ne$

= num of ex charge of ae^-

⑧ Types of element:

- ① Conductor
- ② Semi-Conductor
- ③ Insulator

④ Super Conductors

- ① No resistance
- ② Highest amount of current

Avogadro number = 6.02×10^{23}

$e^- = 1.6 \times 10^{-19} \text{ C}$

Coulomb's law

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$$

$$F = k \frac{q_1 q_2}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$\epsilon_0 \rightarrow$ permittivity Constant

Newton's law

$$F = G \frac{m_1 m_2}{r^2}$$

$\leftarrow \begin{array}{c} + \\ \bullet \end{array} \quad \begin{array}{c} + \\ \bullet \end{array} \rightarrow$ Repulsion

$\leftarrow \begin{array}{c} + \\ \bullet \end{array} \quad \begin{array}{c} - \\ \bullet \end{array} \rightarrow$ Attraction

$\leftarrow \begin{array}{c} - \\ \bullet \end{array} \quad \begin{array}{c} - \\ \bullet \end{array} \rightarrow$ Repulsion

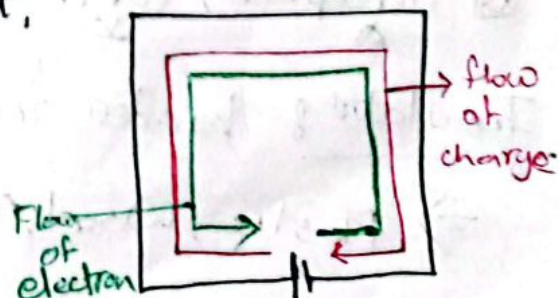
~~Static~~ Electric Charge ~~city~~ pos, neg.

Current: Rate of flow of positive charge. Flow of electrons ions and or other charged particle.

Electrons: Negatively charged. So electrons flow direction

is opposite to current.

electron $\leftarrow$
current $\rightarrow$



So,

$$\text{electric charge (Q)} = \text{electric current (i)} \times \text{time (t)}$$

কি Circuit-এ Positive terminal থেকে Negative
এর দিকে ইলেকট্রিক চার্জের flow করে যা Ground পর্যন্ত,
আমরা তাকে Ground থেকে লক্ষ্য করি।

Static electricity: এমন বিদ্যুতিক চার্জ যা পদার্থ বা পদার্থের
স্বর্ষ থেকে হয়, এই চার্জগুলি স্থিতি থেকে grounded হয়।
Mainly positive এবং Negative চার্জ আলাদা করা হয়ে
থাকে।

$$\text{Number of electron} = \text{Charge/time}$$

Conductors: Materials that allow to flow electricity
is Conductor.

Flow of electrons in conductor $\rightarrow$ electric current
the force required to make that current flow
is Voltage | i.e $\rightarrow$ silver, Gold, Iron, Human body,
Graphite, earth, sea water

Insulator: hinders free flow of electron.

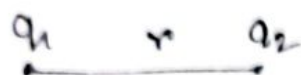
$\Rightarrow$ Plastic, Wood, Glass, Rubber, Foam, Ceramic

Coulomb's law

two charged molberry pithballs repelled each other with a force that is inversely proportional to its distance

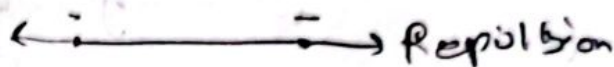
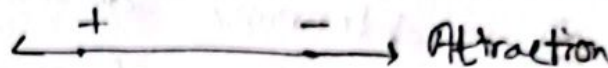
$$F = \frac{k q_1 q_2}{r^2}$$

Where $k = 8.99 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$
 $k \sim 10^{10} \text{ Nm}^2 \text{C}^{-2}$



Point charges

Spheres same as points



The law: The law states that the magnitude, or

absolute value of the attractive or repulsive electrostatic force between two charges is directly proportional to the product of the magnitudes of their charges and inversely proportional to the squared distance between them.

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r} \quad (\text{Coulomb's law})$$

$$\vec{F} = G \frac{m_1 m_2}{r^2} \hat{r} \quad (\text{Newton's law})$$

Gravitational Constant

$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$$

$$8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

permittivity of the area

Coulomb's Constant

electrostatic constant

It holds for stationary charges only which are point sized. [কিছু কিছু আকারের দৃষ্টি গণ্যে এসে শূন্য-মাত্রা] obeys Newton's third law

Application:

- ① To determine d and f between 2 Q's
- ② u u f on one point due to the presence of several points.

Limitations:

- ① The law applies just for the point charges at rest.
- ② Can be just applied ^{these cases where} for inverse squared law is complied with
- ③ Charges remain in approx shape is difficult? Can't identify charge.
- ④ Can't determine charge on the big planets.

Between two electrons separated by a certain distance:
Electrical force / Gravitational force $= 10^{42}$

⇒ u protons u $= 10^{36}$

⇒ u one electron one proton $= 10^{39}$

Relationship between the velocity of light, the permeability of free space and permittivity of free space is given by the expression $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ ✓

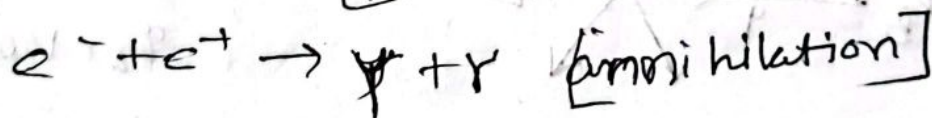
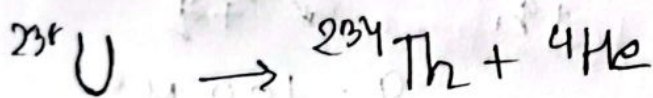
⇒ Electric charge can be quantized

Any (-) or (+) charge q that can be detected written as, $q = ne$; $n = \pm 1, \pm 2, \pm 3, \dots$

where,

e , the elementary charge,

$$e = 1.6 \times 10^{-19} \text{ C}$$

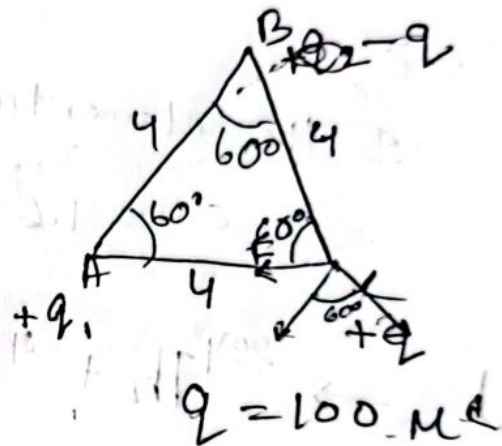
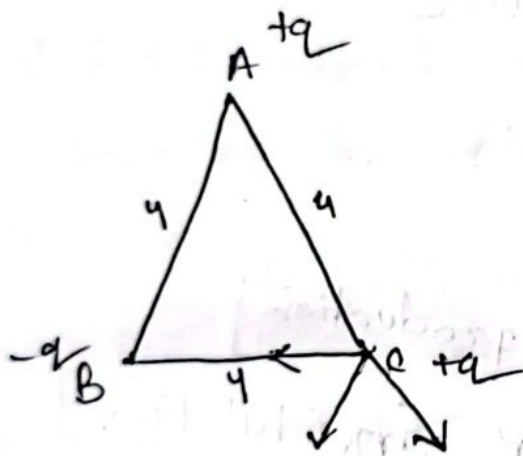


Superposition

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5$$

Problems

Charges of magnitude $100 \text{ micro-}\mu\text{C}$ each are located in vacuum at the corners A, B, C and of an equilateral triangle measuring 4 meters on each side. If the charge at A and C are positive and the charge B neg. what is the magnitude and direction of the total force on the charge C?



$$q = 100 \text{ micro-}\mu\text{C}$$

$$q_1 = 100 \times 10^{-6} \text{ C}$$

$$q_2 = 100 \times 10^{-6} \text{ C}$$

$$q_3 = 10^{-6} \text{ C}$$

$$F_{AC} = k \frac{(100 \times 10^{-6})^2}{4^2}$$

$$= 9 \times 10^9 \frac{(100 \times 10^{-6})^2}{4^2}$$

$$= 5.625 \text{ Newton}$$

$$F_{BC} = -k \frac{(100 \times 10^{-6})^2}{4^2}$$

$$= -5.625 \text{ Newton}$$

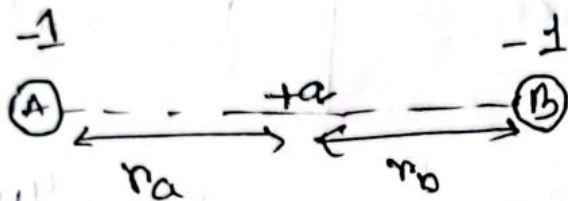
$$F = \sqrt{F_{CA}^2 + F_{BC}^2 + 2F_{CA}F_{BC}\cos 120}$$

$$= 5.625 \text{ newton}$$

AB is parallel to resultant force

Problem 2: The negative point charges of unit magnitude and a positive unit point charge q are placed along the straight line. At what position and for what value of q will the system be in equilibrium. Check whether it is stable, unstable or neutral.

Soln: Let,



$$F_A = k \frac{q \times -1}{(r_a)^2} \quad F_B = k \frac{q \times (-1)}{(r_b)^2}$$

For being in equilibrium both side force would be equal, so,

$$F_A = F_B$$

$$\Rightarrow \frac{k q \times -1}{(r_a)^2} = \frac{k q \times (-1)}{r_b^2}$$

$$\Rightarrow \frac{1}{r_a^2} = \frac{1}{r_b^2}$$

$$\Rightarrow r_a = r_b$$

Q A positive charge of 6×10^{-6} is 0.040 m from the second positive charge of 4×10^{-6} C. Calculate the force between the charges.

Soln:
Here,

$$F = k \times \frac{q_1 \times q_2}{r^2}$$
$$= 9 \times 10^9 \frac{(6 \times 10^{-6})(4 \times 10^{-6})}{(0.040)^2}$$
$$= 135 \text{ N}$$

Given,

$$q_1 = 6 \times 10^{-6} \text{ C}$$

$$r = 0.040 \text{ m}$$

$$q_2 = 4 \times 10^{-6} \text{ C}$$

$$k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

Q Two point charges, $q_1 = +9 \mu\text{C}$ and $q_2 = 4 \mu\text{C}$, are separated by a distance $r = 12 \text{ cm}$. What is the magnitude of the electric force?

Soln:

$$F = k \frac{q_1 \times q_2}{r^2}$$
$$= 9 \times 10^9 \frac{(9 \times 10^{-6})(4 \times 10^{-6})}{(0.12)^2}$$
$$= 22.5 \text{ N}$$

Given,

$$q_1 = +9 \mu\text{C}$$

$$q_1 = 9 \times 10^{-6} \text{ C}$$

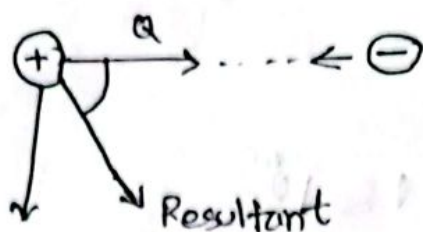
$$q_2 = 4 \mu\text{C}$$

$$= 4 \times 10^{-6} \text{ C}$$

$$r = 12 \text{ cm}$$

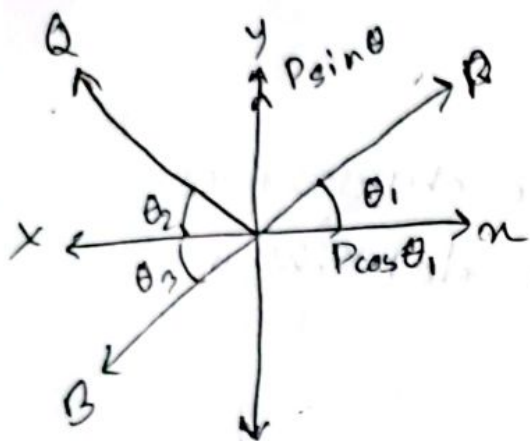
$$= 0.12 \text{ m}$$

change $\rightarrow$



$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$$

$$\alpha = \tan^{-1} \frac{Q \sin \theta}{P + Q \cos \theta}$$



$$R_x = P \cos \theta_1 - Q \cos \theta_2 - B \cos \theta_3$$

$$R_y = P \sin \theta_1 + Q \sin \theta_2 - B \sin \theta_3$$

$$\vec{R} = R_x \hat{i} + R_y \hat{j}$$

$$= (P \cos \theta_1 - Q \cos \theta_2 - B \cos \theta_3) \hat{i} + (P \sin \theta_1 + Q \sin \theta_2 - B \sin \theta_3) \hat{j}$$

$$R = \sqrt{R_x^2 + R_y^2}$$

$$\theta = \tan^{-1} \frac{R_y}{R_x}$$

For 2 protons $F \Rightarrow$

$$\Rightarrow \frac{F_{es}}{F_{gz}} = \frac{k \frac{q_1 q_2}{r^2}}{G \frac{m_1 m_2}{r^2}}$$

$$\Rightarrow \frac{F_{es}}{F_{gz}} = \frac{k (q_1 q_2)^2}{G m_1 m_2}$$

$$\Rightarrow \frac{F_{es}}{F_{gz}} = \frac{k}{G} \left(\frac{q_e^2}{m_p^2} \right)$$

$$= \frac{9 \times 10^9}{6.67 \times 10^{-11}} \left(\frac{(1.6 \times 10^{-19})^2}{(1.67 \times 10^{-27})^2} \right)$$

$$= 1.23 \times 10^{36}$$

$$k = 9 \times 10^9$$

$$G = 6.67 \times 10^{-11}$$

$$q_e = 1.6 \times 10^{-19} \text{ C}$$

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

For $2e^- \rightarrow \approx 10^{42}$

1p and $1e^- \rightarrow \approx 10^{39}$

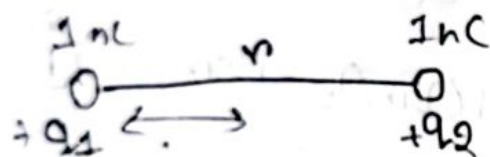
For $2p \rightarrow \approx 10^{36}$

$F_{es} \Rightarrow$ Dominating (Repulsive)

$F_{ez} \Rightarrow$ Dominated (Attracted)

☐ What is the force between two positive charges each 1 nano Coulomb, 1 cm apart in a typical demo? Why is the force so weak here?

Solng



$$F = \frac{k q_1 q_2}{r^2}$$

$$F = \frac{9 \times 10^9 (10^{-9})^2}{(10^{-2})^2} = 9 \times 10^{-5}$$